# AI Personal Finance & Smart Expense Predictor

**Project 03 | Team 03** • **Role:** AI/ML & Expense Prediction • **Author:** Vishwesh Babu T J (with Mitul Jain)

## 1. Executive Summary & Architecture

The AI/ML & Expense Prediction module estimates future monthly spending across categories from historical transaction records, evaluates forecasting accuracy without temporal leakage, and surfaces actionable financial insights. It is designed to help users understand spending habits rather than guarantee future outcomes or provide investment advice.

## 2. Forecasting Methodology & Leakage Prevention

- **Feature Engineering:** Daily expense records are aggregated by `YearMonth` and `Category`. Predictive features include 1-month lag (`Prev_1M_Spend`), 2-month lag (`Prev_2M_Spend`), 3-month rolling average (`Prev_3M_Rolling_Avg`), first-half month spend (Day 1–15), calendar month, and year.
- **Time-Based Evaluation Split:** To prevent random data leakage across time, models are trained strictly on historical records from **2022-04 to 2025-06** (39 months) and evaluated on a held-out future window from **2025-07 to 2025-12** (6 months).
- **Insufficient History Safeguard:** If a user has fewer than 4 months of recorded data (or an empty dataset), the module suppresses ML predictions and returns a clear warning message to avoid false precision.

## 3. Model Evaluation Report (Held-Out Test Period: Jul–Dec 2025)

| Model | Description | Overall MAE (₹) | Relative Improvement vs. Baseline |
|---|---|---|---|
| **Baseline Model** | 3-Month Historical Moving Average | ₹1,170.12 | Baseline Reference |
| **Linear Regression** | OLS Regression on Lag & Temporal Features | ₹1,081.12 | +7.61% lower error |
| **Random Forest Regressor** | Ensemble (150 trees, max_depth=8) | **₹1,050.02** | **+10.26% lower error (Best Overall)** |

### Category-Wise MAE Breakdown (₹)

| Category | Baseline MAE (₹) | Linear Regression MAE (₹) | Random Forest MAE (₹) | Best Model for Category |
|---|---|---|---|---|
| Food | ₹1,590.05 | ₹1,222.03 | ₹1,401.92 | Linear Regression |
| Rent | ₹368.35 | ₹552.50 | ₹604.30 | Baseline (3M Moving Avg) |
| Shopping | ₹2,908.94 | ₹2,691.93 | ₹2,701.73 | Linear Regression / RF |
| Transport | ₹397.81 | ₹466.57 | ₹222.56 | Random Forest |

| Category | Baseline MAE (₹) | Linear Regression MAE (₹) | Random Forest MAE (₹) | Best Model for Category |
|---|---|---|---|---|
| Utilities | ₹585.46 | ₹472.57 | ₹319.58 | Random Forest |

## 4. Data-Driven Financial Insights & Anomaly Detection

Unusual spending spikes are identified when a category's monthly spend exceeds its 6-month rolling mean by more than 2.0 standard deviations ($Z > 2.0$):

- **Festival Shopping Spike (Nov 2025):** Shopping reached **₹13,265.45** (+120.8% above the 6-month rolling average of ₹6,007.26, $Z = 5.26$).
- **Pre-Festival Shopping Increase (Oct 2025):** Shopping reached **₹8,438.87** (+56.1% above the 6-month rolling average of ₹5,404.76, $Z = 4.02$).
- **Food & Dining Surge (Oct 2025):** Food expenses rose to **₹12,644.73** (+62.8% above the 6-month rolling average of ₹7,767.79, $Z = 9.83$).
- **Year-End Transport Increase (Dec 2025):** Transport spend reached **₹4,632.70** (+18.3% above the 6-month rolling average of ₹3,915.72, $Z = 2.05$).

**5. Forecast Assumptions, Limitations & Privacy Note**

• **Assumptions & Limitations:** All forecast outputs are labelled as *Estimates*. Predictions assume historical seasonal and lag relationships continue; sudden life events or unrecorded cash transactions cannot be anticipated.

• **Privacy & Data Handling:** Evaluated entirely on a 4-year synthetic dataset. No real bank credentials, account numbers, or payment card details are collected or stored.